



# Cambridge IGCSE™

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## CHEMISTRY

0620/52

Paper 5 Practical Test

February/March 2026

1 hour 15 minutes

You must answer on the question paper.

You will need: The materials and apparatus listed in the confidential instructions

### INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

### INFORMATION

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [ ].
- Notes for use in qualitative analysis are provided in the question paper.

#### For Examiner's Use

|       |  |
|-------|--|
| 1     |  |
| 2     |  |
| 3     |  |
| Total |  |

This document has **12** pages. Any blank pages are indicated.

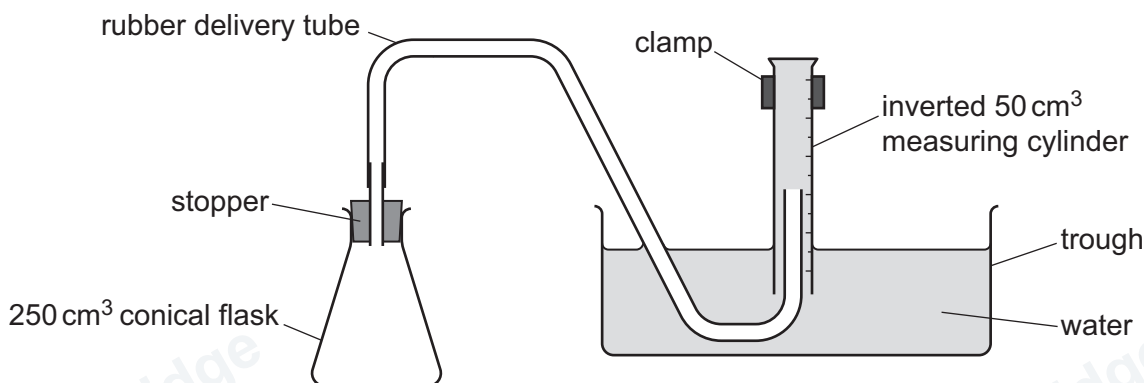
- 1 You are going to investigate the time taken to make  $50\text{ cm}^3$  of carbon dioxide gas when solid sodium hydrogencarbonate reacts with dilute citric acid of different concentrations.



**Read all of the instructions carefully before starting the experiments.**

### Instructions

You are going to do five experiments using the apparatus shown in Figure 1.1.



**Figure 1.1**

#### Experiment 1

- Use a  $50\text{ cm}^3$  measuring cylinder to pour  $50\text{ cm}^3$  of dilute citric acid into the  $250\text{ cm}^3$  conical flask.
- Use the  $100\text{ cm}^3$  measuring cylinder to pour  $55\text{ cm}^3$  of distilled water into the conical flask.
- Swirl the flask to mix the contents.
- Set up the apparatus as shown in Figure 1.1. Ensure the inverted measuring cylinder is full of water.
- Remove the stopper from the conical flask. Ensure the delivery tube remains in the inverted measuring cylinder.
- Add  $1.5\text{ g}$  of solid sodium hydrogencarbonate to the conical flask. Do **not** mix the contents of the flask. Immediately replace the stopper and start the stop-watch.
- Measure the time taken to collect  $50\text{ cm}^3$  of gas in the inverted measuring cylinder. Record the time in seconds, to the nearest whole number, in Table 1.1.
- Rinse the conical flask with distilled water.

#### Experiment 2

- Repeat Experiment 1, using  $50\text{ cm}^3$  of distilled water instead of  $55\text{ cm}^3$ .

#### Experiment 3

- Repeat Experiment 1, using  $45\text{ cm}^3$  of distilled water instead of  $55\text{ cm}^3$ .

#### Experiment 4

- Repeat Experiment 1, using  $40\text{ cm}^3$  of distilled water instead of  $55\text{ cm}^3$ .

#### Experiment 5

- Repeat Experiment 1, using  $30\text{ cm}^3$  of distilled water instead of  $55\text{ cm}^3$ .



(a) Complete Table 1.1.

Table 1.1

| experiment | volume of dilute citric acid / cm <sup>3</sup> | volume of distilled water / cm <sup>3</sup> | time to collect 50 cm <sup>3</sup> of gas / s |
|------------|------------------------------------------------|---------------------------------------------|-----------------------------------------------|
| 1          | 50                                             |                                             |                                               |
| 2          |                                                |                                             |                                               |
| 3          |                                                |                                             |                                               |
| 4          |                                                |                                             |                                               |
| 5          |                                                |                                             |                                               |

[4]

(b) On Figure 1.2, write a suitable scale on the y-axis and plot your results from Experiments 1 to 5.

Draw a line of best fit.

time to collect  
50 cm<sup>3</sup> of gas  
/ s

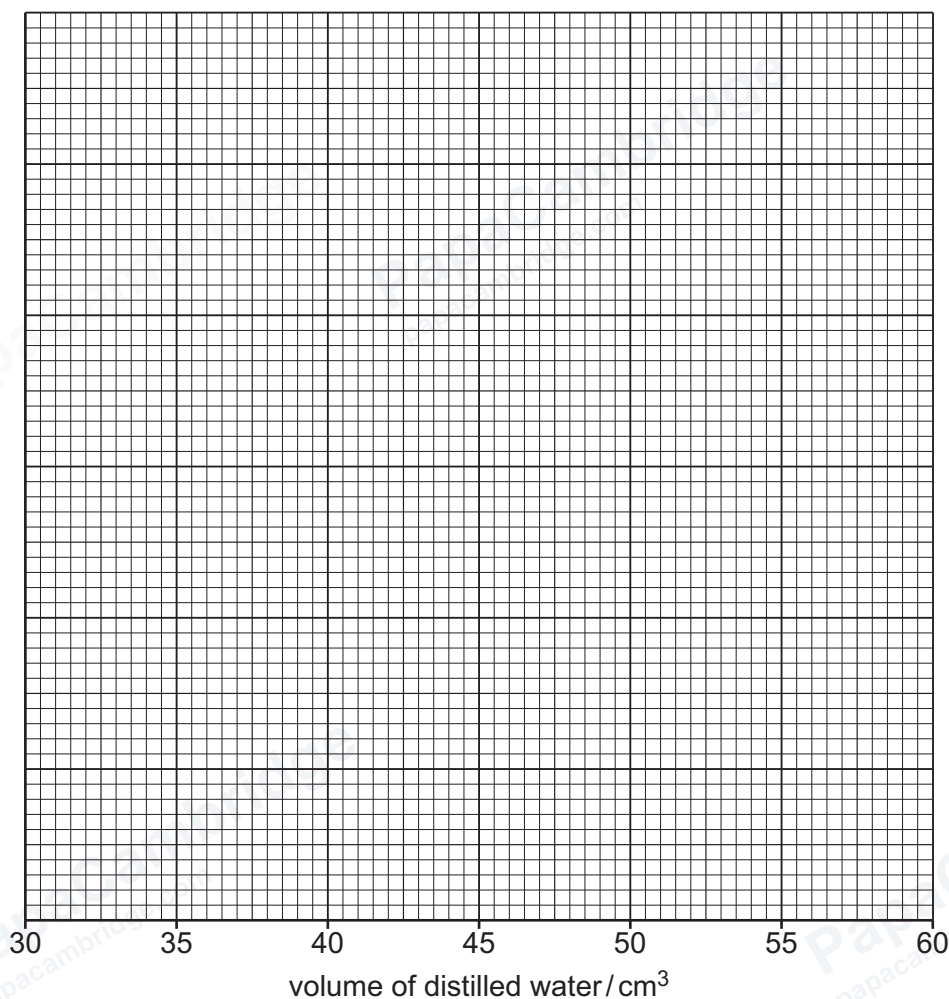


Figure 1.2

[4]

- (c) The mean rate of reaction is calculated using the equation shown.

$$\text{mean rate of reaction} = \frac{\text{volume of gas collected}}{\text{time taken to collect the gas}}$$

- (i) Calculate the mean rate of reaction in Experiment 3.  
Give units for the mean rate you have calculated.

mean rate of reaction = .....

units ..... [2]

- (ii) Use your graph in Figure 1.2 to deduce the time taken to collect 50 cm<sup>3</sup> of gas if 35 cm<sup>3</sup> of water is used.

Show clearly **on Figure 1.2** how you worked out your answer.

time to collect 50 cm<sup>3</sup> of gas = ..... s [2]

- (d) The gas collected in the inverted measuring cylinder contains air from the conical flask.

State what effect, if any, this air will have on the accuracy of the results of the investigation.  
Explain your answer.

effect on accuracy .....

explanation .....

[2]



(e) Some gas escapes in the short time between adding the solid sodium hydrogencarbonate to the conical flask and replacing the stopper in the conical flask.

(i) Explain how the apparatus can be changed so that no gas escapes. You may draw a diagram to help explain your answer.

.....

.....

.....

.....

[2]

(ii) State the effect, if any, the gas escaping has on the time taken to collect 50 cm<sup>3</sup> of gas compared to if the apparatus was changed so that no gas escapes.

..... [1]

(f) On Figure 1.2, sketch the graph you would expect when the investigation is repeated using dilute citric acid of different concentrations at 50 °C instead of at room temperature.

Label this line F.

[1]

[Total: 18]

- 2 You are provided with two substances: solid **A** and solution **B**.

Do the following tests on solid **A** and solution **B**. Record all of your observations at each stage.

**Tests on solid A**

- (a) Transfer about half of solid **A** to a boiling tube. Heat solid **A** strongly in the boiling tube.

Record your observations.

.....

.....

..... [2]

- (b) Transfer the remaining solid **A** to a boiling tube. Add about 5 cm depth of dilute hydrochloric acid to solid **A**. Test any gas given off.

**Keep the solution formed for use in 2(c).**

Record your observations.

.....

.....

..... [2]

- (c) The solution formed in **2(b)** is solution **C**. Carefully pour solution **C** into a boiling tube while leaving the remaining solid behind.

Divide solution **C** into three approximately equal portions in three test-tubes.

- (i) To the first portion of solution **C**, add aqueous sodium hydroxide dropwise and then in excess.

Record your observations.

.....

.....

..... [2]



- (ii) To the second portion of solution **C**, add about 1 cm depth of dilute nitric acid followed by a few drops of aqueous barium nitrate.

Record your observations.

.....  
..... [1]

- (iii) Pour about 2 cm depth of aqueous ammonia into a boiling tube. Add the third portion of solution **C** to the aqueous ammonia in the boiling tube dropwise until it has all been added.

Record your observations.

dropwise .....  
.....  
all of solution **C** added .....  
..... [2]

- (d) Identify solid **A**.

.....  
..... [2]



# Tests on solution B

- (e) Carry out a flame test on solution **B**.

Record your observations.

..... [1]

- (f) Divide the remaining solution **B** into two approximately equal portions in two test-tubes.

- (i) To the first portion of solution **B**, add a few drops of dilute sulfuric acid.

Record your observations.

..... [1]

- (ii) To the second portion of solution **B**, add about 1 cm depth of dilute nitric acid followed by a few drops of aqueous silver nitrate.

Record your observations.

..... [1]

- (g) Identify solution **B**.

..... [2]

[Total: 16]









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## Notes for use in qualitative analysis

## Tests for anions

| anion                                        | test                                                              | test result                                                                             |
|----------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| carbonate, $\text{CO}_3^{2-}$                | add dilute acid, then test for carbon dioxide gas                 | effervescence, carbon dioxide produced                                                  |
| chloride, $\text{Cl}^-$<br>[in solution]     | acidify with dilute nitric acid, then add aqueous silver nitrate  | white ppt.                                                                              |
| bromide, $\text{Br}^-$<br>[in solution]      | acidify with dilute nitric acid, then add aqueous silver nitrate  | cream ppt.                                                                              |
| iodide, $\text{I}^-$<br>[in solution]        | acidify with dilute nitric acid, then add aqueous silver nitrate  | yellow ppt.                                                                             |
| nitrate, $\text{NO}_3^-$<br>[in solution]    | add aqueous sodium hydroxide, then aluminium foil; warm carefully | ammonia produced                                                                        |
| sulfate, $\text{SO}_4^{2-}$<br>[in solution] | acidify with dilute nitric acid, then add aqueous barium nitrate  | white ppt.                                                                              |
| sulfite, $\text{SO}_3^{2-}$                  | add a small volume of acidified aqueous potassium manganate(VII)  | the acidified aqueous potassium manganate(VII) changes colour from purple to colourless |

## Tests for aqueous cations

| cation                          | effect of aqueous sodium hydroxide                                         | effect of aqueous ammonia                                                  |
|---------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------|
| aluminium, $\text{Al}^{3+}$     | white ppt., soluble in excess, giving a colourless solution                | white ppt., insoluble in excess                                            |
| ammonium, $\text{NH}_4^+$       | ammonia produced on warming                                                | —                                                                          |
| calcium, $\text{Ca}^{2+}$       | white ppt., insoluble in excess                                            | no ppt. or very slight white ppt.                                          |
| chromium(III), $\text{Cr}^{3+}$ | green ppt., soluble in excess                                              | green ppt., insoluble in excess                                            |
| copper(II), $\text{Cu}^{2+}$    | light blue ppt., insoluble in excess                                       | light blue ppt., soluble in excess, giving a dark blue solution            |
| iron(II), $\text{Fe}^{2+}$      | green ppt., insoluble in excess, ppt. turns brown near surface on standing | green ppt., insoluble in excess, ppt. turns brown near surface on standing |
| iron(III), $\text{Fe}^{3+}$     | red-brown ppt., insoluble in excess                                        | red-brown ppt., insoluble in excess                                        |
| zinc, $\text{Zn}^{2+}$          | white ppt., soluble in excess, giving a colourless solution                | white ppt., soluble in excess, giving a colourless solution                |



**Tests for gases**

| gas                           | test and test result                                                       |
|-------------------------------|----------------------------------------------------------------------------|
| ammonia, $\text{NH}_3$        | turns damp red litmus paper blue                                           |
| carbon dioxide, $\text{CO}_2$ | turns limewater milky                                                      |
| chlorine, $\text{Cl}_2$       | bleaches damp litmus paper                                                 |
| hydrogen, $\text{H}_2$        | 'pops' with a lighted splint                                               |
| oxygen, $\text{O}_2$          | relights a glowing splint                                                  |
| sulfur dioxide, $\text{SO}_2$ | turns acidified aqueous potassium manganate(VII) from purple to colourless |

**Flame tests for metal ions**

| metal ion                    | flame colour |
|------------------------------|--------------|
| lithium, $\text{Li}^+$       | red          |
| sodium, $\text{Na}^+$        | yellow       |
| potassium, $\text{K}^+$      | lilac        |
| calcium, $\text{Ca}^{2+}$    | orange-red   |
| barium, $\text{Ba}^{2+}$     | light green  |
| copper(II), $\text{Cu}^{2+}$ | blue-green   |

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